Project Proposal: Generative AI Capstone

IS Professional Elective 4 — Generative AI Systems

Instructor: Jessie Melendres

1. Project Identity

Project Title: FitGen: A RAG-based Telegram Chatbot for Personalized Workout Plan Generation

Chosen Theme: Others (please specify): Personal Fitness / Health & Wellness Assistant (deployed as a Telegram bot)

Repository Link: https://github.com/angeloxparedes-cloud/chatbot

2. Problem Statement

Many individuals want to start or improve a fitness routine but don't know how to structure a workout plan that fits their specific goals, experience level, available equipment, and schedule. Generic workout plans found online are often one-size-fits-all, not tailored to the user's fitness level or constraints, and can even pose injury risk for beginners if exercises are improperly sequenced or too advanced.

Hiring a personal trainer is effective but costly and inaccessible for many, especially students and budget-conscious individuals. Free fitness apps and static workout PDFs lack the flexibility to answer follow-up questions, adjust plans based on user feedback (e.g., “this exercise hurts my knees,” “I only have dumbbells”), or explain the reasoning behind exercise selection. Delivering this through Telegram — a platform most users already have installed — removes the friction of downloading a separate app, making personalized fitness guidance instantly accessible through a familiar chat interface.

3. Target Users

Primary users: Individuals seeking a personalized workout routine — beginners unsure where to start, and intermediate users wanting structured progression, without access to a personal trainer.

Secondary users: Fitness enthusiasts or students who want a quick, chat-based reference tool to generate or adjust routines based on available equipment, time, or muscle group focus.

This tool shifts users from manually researching and piecing together workout advice from scattered sources, to messaging a Telegram bot and instantly receiving a structured, sourced routine tailored to their goals — with the ability to ask follow-up questions (e.g., substitutions, form tips) in natural conversation.

4. Dataset Source(s)

Data Type: Markdown and CSV documents (exercise guides, training principles, structured exercise database).

Source: Publicly available exercise science references and fitness guidelines (e.g., ACE/ACSM-style training principles, reputable fitness education sites), open exercise databases (e.g., free exercise datasets listing muscle groups, equipment, and difficulty level), and compiled reference sheets on workout programming principles (sets/reps schemes, progressive overload, rest periods, beginner vs. advanced splits).

Volume (as implemented): 57 source documents — 52 exercises in a structured CSV (name, muscle group, equipment, difficulty, movement pattern, instructions, precautions) and 5 markdown reference guides (progressive overload, warm-up/cool-down, program design & splits, injury safety, sets/reps/rest guidelines), yielding 84 indexed chunks after preprocessing.

Note: the current dataset is a realistic, instructor-approved sample built to match the scope and structure described above for development and testing. It can be swapped for a fully source-verified dataset (e.g., specific cited ACE/ACSM pages or a named open exercise dataset) before final submission if required.

5. Success Criteria

- Technical Accuracy: The system correctly cites its source/reference for at least 90% of exercise recommendations and training principles it states.
- Performance: Average retrieval + response latency is under 3 seconds per query within the Telegram chat interface.
- Utility: Successfully generates complete, coherent workout plans for at least 5 distinct user profiles (varying goal, experience level, equipment access) without hallucinating unsafe or contradictory advice.
- Personalization: The generated plan correctly reflects all stated user constraints (e.g., excludes exercises requiring unavailable equipment, respects stated injury limitations) in at least 90% of test cases.
- Usability: At least 3 test users interact with the live Telegram bot and rate the generated workout plans as "clear and appropriately tailored" in a post-test survey.

6. Team Composition

As per course requirements, each teammate must own a clear pipeline component.

Team Member Name	Primary Responsibility	Assigned Component
Angel Frederick Paredes	Lead Developer	Data Pipeline & Embeddings, RAG Architecture & Prompting, Telegram Bot Deployment & Interface, Testing, Evaluation & Technical Documentation
Jeeruz Marl Tabar	Research Support	Compiled exercise/training reference sources; reviewed prompt outputs
Luke Kibry Jao	Testing Support	Manual end-to-end testing of the Telegram bot across sample profiles
Clein Lord Will Surigao	Documentation Review	Reviewed checkpoint write-ups and the final report